
Evaluating the total effect of early supplier involvement on project team effectiveness: collaboration and interaction

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Abstract: Motivated by the inconsistent findings of the linkage between early supplier involvement practice and new product development success, this study adopts both the resource-based perspective and agency theory to investigate dynamic relationships between the manufacturer and supplier during the collaboration process. Survey data were used to empirically examine the impact of the resource factor (i.e. collaboration effect) and agency factors (i.e. interaction effect) on project team effectiveness. The results showed that selecting suppliers with appropriate knowledge and resources is necessary yet insufficient to improve project team effectiveness. The agency problem plays a dominant role in determining the quality of collaboration. In particular, the study identified information sharing and task programmability as two key governance mechanisms to curb suppliers' opportunistic behaviour. The study provides insightful managerial and policy implications for firms considering early supplier involvement practice in their innovation process.

Keywords: early supplier involvement; new product development; agency theory; collaboration; interaction.

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1 Introduction

In the turbulent and vibrant environment where technical and market changes can never be fully controlled, proactive New Product Development (NPD) has become a potential source of dynamic competitive advantage (Clark and Fujimoto, 1991; Brown and Eisenhardt, 1995; van Echtelt et al., 2008). In industries ranging from software to automobiles, firms that develop their own products can rapidly gain the attention of customers and expand their market share significantly.

While Original Equipment Manufacturers (OEMs) increasingly outsource parts and components production to suppliers to gain production efficiency, they face major challenges in product innovation including the loss of hierarchical control to their partners and the lack of knowledge and expertise in specific areas. As such, the locus of innovation has shifted from single firm to groups of networked firms (Lin and Chen, 2006). Inter-firm collaboration becomes a viable path to providing an efficient mechanism to leverage the supplier's Research and Development (R&D) capability in the NPD process. The emergence of Early Supplier Involvement (ESI) practice in NPD and its implementation in many OEMs as well as outsourced innovations reflect this trend (Quinn, 2000).

ESI is a form of vertical cooperation in which manufacturers involve suppliers at an early stage in the product development/innovation process. Typically, this occurs at the level of concept and design (Bidault et al., 1998; LaBahn and Krapfel, 2000). By practicing ESI, Japanese manufacturers are able to bring new automobiles to market faster, with more innovative features but less development effort in terms of number of hours and engineers (Wynstra et al., 2001). The practice has been widely adopted in industries such as aerospace, automotive, mainframe and personal computers (Sobrero and Roberts, 2002), spurring a simultaneous increase in academic interest in ESI.

While earlier and more extensive involvement of suppliers in NPD is argued to be one of the ways to enhance product development performance in terms of productivity, speed, and/or product quality (Droge et al., 1999; Primo and Amundson, 2002; Ragatz et al., 2002; Lyu and Chang, 2007), several studies also demonstrated that managing supplier involvement in product development poses quite some challenges (Birou and Fawcett, 1994; Hartley et al., 1997; Mikkola and Skjott-Larsen, 2003; Petersen et al., 2005). Numerous empirical studies conducted to assess the impact of ESI on NPD performance or new product performance but only found inconsistent results. Considering the fact that many firms attempt to leverage the capabilities of their supply partners to develop new products, there is a critical need to explore the problems in practicing ESI, therefore enable them to extract maximum benefits from inter-firm collaboration.

The purpose of this study is to empirically evaluate the overall effect of ESI on new product development team's effectiveness. We achieved the objective by synthesising two theories – the resource-based view and the agency theory. While the resource-based view explains the positive collaboration effect, the agency theory serves as the foundation for examining the potential interaction effect between the manufacturer and supplier. Though plausible in theory, there has not been any empirical study that assessed the overall effect and the relative importance of these two effects. Key findings from this study offer the management team of the manufacturer insights of how best to utilise the key suppliers' resources and capabilities.

The rest of the paper is organised as follows. Section 2 reviews the literature regarding the impact of ESI on NPD. Section 3 introduces two relevant theoretical frameworks pertaining to ESI and presents the conceptual model and associated hypotheses for testing. The methodology including data collection and construct measurement is discussed in the next section. The major findings are presented in Section 5, followed by discussions of the findings. The last section offers conclusions, limitations and directions for future research.

2 Literature review

There has formed a strong stream of literature on ESI since 1990s. To serve for the purpose of this study, the literature review focuses on the impact of ESI on new product performance or NPD performance. A review of this group of literature reveals contradictory results. On the one hand, many empirical studies have presented positive benefits of process efficiency and product effectiveness including cost reduction, quality improvement and shorter development cycle (Birou and Fawcett, 1994; Ragatz et al., 1997; Droge et al., 1999; Wynstra et al., 2001; Lyu and Chang, 2007). On the other hand, some studies have concluded that ESI has failed to live up to the expectation (Eisenhardt and Tabrizi, 1994; Zirger and Harley, 1996; Von Corswant and Tunalv, 2002). For instance, Littler et al. (1995) found that collaboration with suppliers in the NPD process is more costly and slower in the information and communication technology sector because suppliers make NPD projects more complex. Mikkola and Skjoett-Larsen (2003) pointed out that suppliers also introduce serious problems by aggravating the complexity of managing NPD projects. Despite positive buyers' perception about supplier's contribution to product development, Hartley et al. (1997) found that supplier involvement had little practical influence on overall project technical success. ESI was also found to have no significant impact on product development time in the electronics industry (Zirger and Harley, 1996). Eisenhardt and Tabrizi (1994) uncovered that supplier involvement, in general, has no impact on high uncertainty industries such as computer industry. In a survey study only 20% of the respondents agreed that they were satisfied with the results of supplier integration efforts while over 45% disagreed (Handfield et al., 1999). These inconsistent results become the major thrust for this study.

Recent studies have discussed two different ways to possibly resolve these inconsistent results (Monczka et al., 2000; Wynstra et al., 2001; Primo and Amundson, 2002; Wagner and Hoegl, 2006). The first is to consider contingency factors at the organisation level; and the second is to study the coordination of collaboration. The first implies that ESI is not a 'one-size-fits-all' practice that applies to every circumstance (Kamath and Liker, 1994; Monczka et al., 2000; Laseter and Ramdas, 2002). The second concerns potential interaction between the two parties during collaboration. As pointed out by McIvor et al. (2006), the literature on ESI in the design process typically focuses on outputs and neglects

the actual dynamics and factors that influence the process of supplier integration. While most studies emphasise the collaboration effect, few address the collaboration governance and interaction between the manufacturer and supplier (Gerwin, 2004).

This study follows the second line and advances the existing studies by offering theoretical foundation to examine both the collaboration effect and interaction effect where the latter reflects the quality of collaboration. This study makes at least three contributions to the body of knowledge of ESI. First, the current literature is short of theoretical argument for the positive or negative impact of ESI on NPD. The majority of the arguments or conclusions are based on industrial observations and case studies/interviews (Mikkola and Skjott-Larsen, 2003; Lyu and Chang, 2007); and most studies are still in the nature of exploratory. For instance, Jayaram's (2008) listed a broad array of supplier integration practices affecting NPD project performance from the literature. Then through an exploratory factor analysis he verified that all these practices could be grouped into three factors that are related to one or more measures of new product development project performance. We advanced prior studies by proposing theoretical frameworks to explain firms' differing NPD project performance. The primary objective is to confirm the hypotheses derived from the theories and examine the applicability of these theories in the context of ESI.

Second, while the literature has focused on the collaborative effect of using suppliers in the design stage, few took into account of the supplier's probability of obstructing in NPD (Primo and Amundson, 2002). Therefore, most empirical models suffer from missing important variables. Flynn et al. (2000) noticed that suppliers' lack of priority and cooperation (i.e. supplier obstructionism) could cause damage to NPD due to dependence on powerful yet uncooperative suppliers. Primo and Amundson (2002) further argued that supplier obstructionism is not the opposite of supplier involvement, and it may occur simultaneously with high supplier involvement. As such, effort is required to integrate suppliers into the NPD process (Susman and Ray, 1999); and the additional coordination time might offset the gains in lead-time from supplier involvement (Clark, 1989). Grounded on the resource-based view and agency theory, this study considers both the positive collaborative effect and the negative interaction to capture the overall dynamics of the supplier involvement process and further understand what makes for success in ESI.

Third, this study uses secondary survey data to assess the dominance of the collaboration effect and interaction effect in the involvement process. The sample is well spanned from small-sized to large-sized firms, across a wide range of industries. As such, the results obtained from this sample give us confidence for generalisation (with caution). That is, the findings are not limited to a specific situation and could offer a general guideline to firms considering using suppliers to facilitate the new product design process.

3 Conceptual model and hypotheses

The theoretical foundations of this study are introduced in this section with a theoretical model and associated hypotheses proposed afterwards.

3.1 Collaboration effect from the resource-based view

ESI is a means of integrating suppliers' capability in the manufacturer's supply chain system and operations (Dowlatshahi, 1998). Suppliers in particular are often capable of not only contributing sound product ideas but also offering creative suggestions on the

technical means to realise them (McGinnis and Vallopra, 1999; Harmsen et al., 2000). Contemporary concurrent engineering doctrine promotes building competitive advantage through the inclusion of members from every functional area (Atuahene-Gime and Evangelista, 2000); the same logic can be applied to include members from outside the firm. At the heart of this argument is the Resource-Based View (RBV) of the firm (Wernerfelt, 1984; Barney, 1991) that believes a firm's heterogeneous resources determine its competitiveness in the market. Resource bundles remain heterogeneous due to following reasons:

- 1 resource market imperfections characterised as barriers to acquisition, imitation, substitution of key resources (Penrose, 1959; Barney, 1991; Schoemaker and Amit, 1994)
- 2 imperfect imitability of firm-specific assets that are accumulated internally over time (Rumelt, 1984; Dierickx and Cool, 1989).

Such resource bundle serves as an enduring source of competitive advantage.

From the resource perspective, the efficiency of activities is contingent on the way resources are utilised and deployed. Available resources can be combined in new ways, introducing dynamics and innovation. Kleinschmidt et al. (2007) used the resource-based theory to argue that four organisational resources are relevant for building global NPD process capacity which is the primary determinant of competitive advantage. Resources are not only those residing inside the boundary of the firm. Supplier involvement has made access to the resources of other firms as important as the resources acquired and deployed inside the firm (Gadde and Hakansson, 2001). Supplier involvement implies the combining of the manufacturer's and supplier's R&D resources and the exploitation of joint capabilities through the strategic relationship (Wagner and Hoegl, 2006). That is, ESI provides the manufacturer not only the access to the supplier's complementary resources, but also an opportunity to form desired capabilities by the integration of resources across firms (Wynstra et al., 2001; Ragatz et al., 2002). This eventually shapes the resource bundle of the manufacturer over time, making it more valuable, inimitable and non-substitutable. As such, positive effects are expected through this collaborative effect.

3.2 Interaction effect from the agency theory

It has to be noted that there actually exists an implicit assumption for ESI to bring desired benefits to the manufacturer from the resource perspective. That is, the supplier's R&D resources (e.g. expertise, know-how or knowledge) can be effectively utilised in the collaboration process. In fact, the total effect of the supplier's resources depends on both the amount of resources (i.e. quantity) and the productivity of resources (i.e. quality). Consequently, the integration of the suppliers' R&D resources refers not only to the access to complementary resources and the expansion of the resource bundle but also the more efficient utilisation of these resources. The quality side of supplier integration, which is the result of interaction between the two collaboration parties, has not been recognised until recently (Hoegl and Wagner, 2005; Dayan and Di Benedetto, 2009).

Lam et al. (2007) argued that manufacturers and suppliers are dependent on each other in collaborative NPD. The interdependence and differences between them (they differ in many aspects such as objective, culture and belief) could lead to conflict throughout collaboration. Conflict, although proved to be critical at an intra-organisational level (Dyer and Song, 1998; Gobeli et al., 1998; Xie et al., 1998), has not been well studied in manufacturer-supplier collaborative NPD context. Conflict intensity has been found to

unfavourably influence NPD performance (Gobeli et al., 1998; Lam et al., 2007). Given the dynamic relationship between the manufacturer and supplier, their interaction becomes another critical factor to understanding ESI.

In order to understand the interaction effect, we have to observe the behaviour of the two parties involved. This can be best positioned in the context of agency theory wherein the principal (e.g. the manufacturer) delegates work to an agent (e.g. the supplier). An agency problem typically arises when the desires or goals of the principal and the agent conflict, and/or, when it is difficult or expensive for the principal to verify what the agent is actually doing (Jensen and Meckling, 1976; Eisenhardt, 1989). Consequently, the agent may behave opportunistically in its own interest and limit its effort towards fulfilling the agreement.

Eisenhardt (1989) argued that the agency theory is most relevant in situations in which contracting problems are difficult. These include situations in which there is (a) substantial goal conflict such that agent opportunism is likely (e.g. buyer–supplier relationship); (b) sufficient outcome uncertainty (e.g. product innovation) and (c) unprogrammed or team-oriented jobs in which evaluation of behaviour is difficult. Apparently, the agency theory provides a great leverage in analysing the manufacturer and supplier's collaboration in ESI.

First, the principal and the agent may have different objectives and preference, primarily because they are financially independent (Eliashberg and Donald, 1984; Logan, 2000). The NPD project could be critical to the manufacturer but not to the supplier. After all, this manufacturer is only one of the many customers of the supplier and thus, the supplier needs to allocate resources to different projects and reduce project specific risk. Second, being an innovation process, it is difficult to predict the outcome and write an explicit contract based upon potential outcomes. This gives a higher degree of freedom to the supplier to decide what they do and how much they do. For instance, while the manufacturer prefers a radical innovation, the supplier may end up delivering an incrementally innovated product, considering the high effort level and the little chance of success. Finally, it is also difficult or costly for the principal to measure or monitor the performance of the agent (Poppo and Zenger, 1998). This is particularly true in a team-oriented work such as ESI in which many spontaneous activities or initiatives are required along the way and, it is impossible to specify all the activities at the outset as a reference point for evaluation. Additionally, the manufacturer may not have sufficient capability to appropriately evaluate the supplier's productivity, innovativeness and progress on the project as it loses the exact expertise in the specific areas.

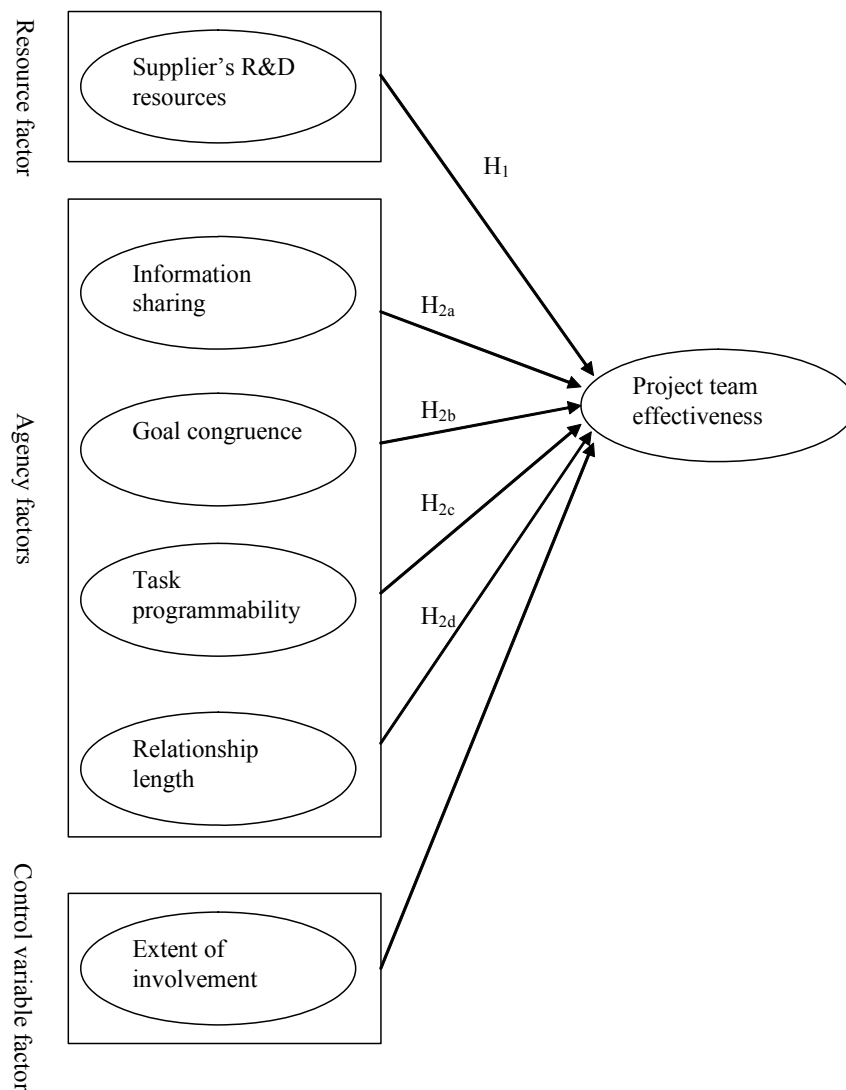
Due to these reasons, the supplier may have both the motive and opportunity to take advantage of the manufacturer and deviate from what it is supposed to do and deliver. Once the supplier does not work in the best interest of the manufacturer, the friction reduces the quality of collaboration and eventually negatively influences NPD performance.

3.3 Conceptual model and hypotheses

Based upon the prior discussion, a conceptual model (Figure 1) is proposed to reflect the two forces in ESI. The agency theory not only suggests choice of 'contract' but also proposes governance mechanisms that could limit the agent's self-serving behaviour. After reviewing the key literature of the agency theory, Zsidisin and Smith (2005) summarised eight agency factors as safeguarding mechanisms – information sharing, programmability, relationship length, goal congruency, adverse selection, moral hazard, outcome uncertainty and risk aversion. In our research model, we exclude the last four factors due to the following reasons.

First, adverse selection is not a substantial issue in ESI because the manufacturer typically knows its key suppliers' technical capabilities very well and also goes through a thorough selection process to decide whom to collaborate with. That is, the chance is very slim for the manufacturer to choose an incompetent supplier. Second, moral hazard cannot be operationalised objectively since it is difficult for the manufacturer to give a fair evaluation on the supplier effort. Third, it is reasonable to assume that both parties have a similar attitude towards risk since the literature shows that the primary purpose of using suppliers is to leverage their capabilities rather than to avoid or shift risk (Wynstra et al., 2001; Ragatz et al., 2002). Lastly, outcome uncertainty is an inherent characteristic of product innovation. Reducing it means it is unlikely to generate highly innovative products. As such, the manufacturer cannot do much to alleviate the agency problem derived from this factor but to accept it.

Figure 1 Proposed research model



In summary, we only retain those agency factors that the manufacturer is able to influence the supplier's behaviour in the conceptual framework. Along the supplier involvement process, asymmetric information and goal misalignment are the major issues that influence the interaction between the manufacturer and the supplier. The manufacturer can either reduce the information asymmetry by facilitating information sharing to prevent the agent from 'shrinking' (Fama, 1980), or structure the incentives to co-align the two parties' interests (Levinthal, 1988; Bergen et al., 1992). To capture the interaction between the two parties, we used an NPD project as our unit of analysis and observed the effects on project team effectiveness from the resource perspective and the agency theory.

H₁: The supplier's R&D resources are positively associated with project team effectiveness.

H₁ describes the direct effect from the resource perspective when suppliers are involved in the development team. With complementary knowledge and resources from the supplier, the resource bundle of the manufacturer changes and so does its R&D capabilities. The new product development team will work more efficiently simply from the span of the knowledge. Therefore, the team will function more smoothly, make better and faster decisions, improve team consensus and set more realistic overall objectives.

The next set of hypotheses (H_{2a}–H_{2d}) considers four agency factors that could also impact the dynamic behaviours of both parties and therefore project team effectiveness.

H_{2a}: Highly effective information sharing between the manufacturer and supplier is positively associated with project team effectiveness.

There exists asymmetric information between the manufacturer and the supplier – the manufacturer has more knowledge about customers' expectation on the overall new product whereas the supplier has more information about its specific R&D capabilities. The manufacturer's information sharing facilitates the development of trust and promotes goal alignment between the two. On the other side, the supplier's information sharing about its cost structure and technology makes the manufacturer relatively easier to monitor and evaluate the supplier's behaviour and performance. Overall, investing in information systems and strengthening the effective communication between the two parties help the development team function like an integral unit and make decisions more effectively, as suggested by H_{2a}.

H_{2b}: Goal congruence between the manufacturer and supplier is positively associated with project team effectiveness.

When the agent's goal is tightly tied with that of the principal, the agent will behave as the principal would regardless of whether his behaviour is monitored or not. For instance, if the project with the manufacturer could generate significant profit or market share for the supplier, the supplier's best interest would be quite consistent with that of the manufacturer. Therefore, H_{2b} argues that the supplier will assume its responsibility to work whole-heartedly to contribute its knowledge to product design and development.

H_{2c}: Task programmability is positively associated with project team effectiveness.

Task programmability refers to the degree to which appropriate behaviour by the agent can be specified in advance (Eisenhardt, 1989). The more programmed the task, the easier to observe the agent behaviour. The extreme case is the complete information

scenario in which case agency problems will not occur. In the collaboration process, if the manufacturer understands its responsibility, its supplier's responsibility, and the steps/activities the supplier needs to go through to fulfil the responsibility, there is less room for the supplier to hide critical information or even be deceptive with the manufacturer. Consequently, H_{2c} proposes that task programmability enhances team work effectiveness.

H_{2d}: Longer collaboration relationship between the manufacturer and supplier is positively associated with project team effectiveness.

Lastly, when the principal maintains a longer collaboration with the agent, the agent is likely to behave in the interest of the principal. Dowlatshahi (1998) underscored that the relationship between the supplier and the manufacturer is crucial in the implementation of ESI. H_{2d} argues that since a longer relationship proves a successful collaboration of mutual benefits to each party, trust is nurtured and developed between the two organisations. This serves as a substitute for a formal contract to govern the behaviours of both parties and align their interests together.

Essentially, the conceptual model argues that not only the supplier's complementary R&D resources but also appropriate governance structures are necessary to improve new product development team effectiveness. Once effectiveness is improved, better product design as well as improved financial performance of the manufacturer can be expected (Petersen et al., 2005).

In addition to the key factors in Figure 1, many researchers have also recognised that the extent of supplier involvement is relevant to the success of ESI (Kamath and Liker, 1994; Monczka et al., 2000; Laseter and Ramdas, 2002; Parker et al., 2008). Though important, this factor is not the core of this study. While controlling its effect, this study aims to evaluate the relative importance of the resource factor and agency factors. If its main effect turns out to be significant, we will conduct *post hoc* moderating effect analysis to further explore its interaction with the resource factor as well as agency factors.

4 Research methodology

4.1 Data collection

Survey data were used to test the research model. The unit of analysis in this study is a specific case of supplier involvement in the NPD process. Respondents were asked to select a completed case of supplier involvement with which they were familiar and answer the questions based on the experience in that particular project. The primary sample was drawn from purchasing/sourcing managers and executives.

The survey was developed, pre-tested and reviewed by a team of industry experts and a team of academics from organisational behaviour, marketing, operations management and purchasing disciplines. A mailing list of approximately 3000 companies' contacts was obtained from a third party. An initial two-page commitment letter and a fax response form were mailed to them, and of these, 225 indicated that they would be willing to participate. Surveys were mailed to the 225 companies and 134 responses were

received with a response rate of 60.44%. The responding companies represented a wide range of industry groups such as aerospace, automotive, chemicals, computers and electronics, consumer products, industrial equipment, medical products and services, process industries and telecommunications. The companies' sales (in US dollars) ranged from \$3 million to \$160 billion with a median of \$3.1 billion. A majority of the responses (68%) came from North American companies.

Ten out of the 134 responses were excluded from further consideration primarily because of a high proportion of missing values. The random missing values in other responses were replaced by mean values. Responses of the early respondents and late respondents were compared under the assumption that late respondents act more similarly as non-respondents (Armstrong and Overton, 1977). All the paired-sample *t*-test results showed no significant difference between the group of the first 20 respondents and the group of the last 20 respondents, indicating that non-response bias did not seem to affect the results.

4.2 Construct measurement

The extent of involvement is frequently conceptualised as the level of responsibility that the supplier takes in the whole development project (Handfield et al., 1999; Monczka et al., 2000). In the survey, respondents were asked to check the extent of involvement from three choices. The first choice represents the entering of manufacturer and supplier into an informal, or sometimes a formal joint development effort, which may include information and technology sharing and joint decision-making regarding design specifications. The second choice implies that the supplier is informed of customer requirements and is also given almost complete responsibility for the purchased item, with only review and concurrence on the purchased item's specifications by the buying company. The third choice indicates the highest level of involvement, where the supplier first helps identify customers' requirements and generates new product ideas, and is then given almost complete responsibility for the procured items, with only review and concurrence on the purchased item's specifications by the buying company. As such, the extent of involvement is a categorical variable in the statistical model.

All the other constructs in Figure 1 are multi-item latent variables. Each item is measured by a 7-point Likert scale. In order to test the paths among the constructs, the psychometric properties of the latent constructs were assessed. Construct validity (i.e. convergent validity and discriminant validity) was evaluated by confirmatory factor analysis and the reliability of each construct was assessed by Cronbach's α . Table 1 shows the relevant scale description and reports the statistics. All the standardised factor loadings were above 0.54 and significant at $p < 0.01$, providing sufficient evidence of convergent validity. Discriminant validity was specifically assessed by using two-factor confirmatory factor analysis models involving each possible pair of the constructs, with the correlation between them first set free and then constrained to one (Bagozzi and Yi, 1991). In all cases, the χ^2 value of the unconstrained model was significantly lower than that of the constrained one, indicating strong discriminant validity among all the constructs. All the multi-item measures had Cronbach's α greater than 0.70, which suggests adequate reliability (Cook and Campbell, 1979).

Table 1 Construct and measurement statistics

Constructs	Items	Mean	St. dev	Std. loading	Cronbach's α
Supplier's R&D resources	Please rate the importance of each of the following factors in selecting the specific supplier who was integrated in this NPD effort.				
	SR1	2.32	1.27	0.78	0.71
	SR2	2.21	1.25	0.61	
	SR3	2.80	1.52	0.62	
Information sharing	Please rate how effectively each of the following practices was used in this supplier integration effort.				
	IS1	3.16	1.37	0.57	0.73
	IS2	2.54	1.23	0.88	
	IS3	2.27	1.25	0.66	
Goal congruence	Please rate the extent to which you either agree or disagree with the following statement.				
	GA1	2.53	1.19	0.54	0.79
	GA2	2.09	1.09	0.85	
	GA3	1.96	1.17	0.86	
Task programmability	Please rate the extent to which you either agree or disagree with the following statement.				
	PG1	2.27	1.11	0.56	0.87
	PG2	2.34	1.17	0.84	
	PG3	2.25	1.13	0.92	

Table 1 Construct and measurement statistics (continued)

Constructs	Items	Mean	St. dev	Std. loading	Cronbach's α
<i>Relationship length</i>					
<i>Please rate the extent to which you either agree or disagree with the following statement.</i>					
RL1	Our business unit had significant previous business experience with this suppliers	2.81	1.95	0.72	0.85
RL2	Our business unit had significant previous new product development integration experience with this supplier.	3.93	1.97	0.94	
<i>Project team effectiveness</i>					
<i>Please rate the extent to which you either agree or disagree with the following statement.</i>					
TE1	My business unit's project team functioned more smoothly as a result of supplier involvement in the design process.	2.41	1.23	0.82	0.85
TE2	The supplier's involvement in this project led to better decisions by the project team.	2.24	1.18	0.86	
TE3	The supplier's involvement in this project led to faster problem solution during the designing process.	2.13	1.11	0.72	
TE4	The supplier's involvement in this project led to greater consensus on decisions made by the project team.	2.56	1.15	0.73	

Notes: All the factor loadings are significant at $p < 0.01$.

$\chi^2 = 188.90$ with 121 degree of freedom. Bentler-Bonett normed fit index = 0.92; comparative fit index = 0.97; Bollen's fit index = 0.97; root mean-square error of approximation = 0.08.

Since both the predictor and criterion variables were collected from a single respondent of a NPD project at the same time, the study was exposed to common method bias which could either inflate or deflate observed relationships between constructs (Podsakoff et al., 2003). To assess the presence of common method bias, Harman's single-factor test was applied in a confirmatory factor analysis where all the measurements were loaded on one latent variable. This model generated $\chi^2 = 780.67$ with $df = 136$. Comparing this χ^2 result with the prior one ($\chi^2 = 188.90$ with $df = 121$, in the note of Table 1), we rejected the hypothesis of the presence of a single factor accounting for the majority of variance among the measures. Therefore, common method bias did not appear to pose significant problems in this study. Overall, the constructs exhibited good measurement properties.

5 Major findings

Since the items for each construct had great convergent validity and they measured a single construct, the factor score of each construct was extracted and used in the next stage regression analysis in SPSS. Table 2 reports the descriptive statistics of all the constructs in Figure 1.

Table 2 Descriptive statistics of constructs

	<i>SR</i>	<i>IS</i>	<i>PG</i>	<i>RL</i>	<i>GA</i>	<i>Extent</i>	<i>TE</i>
SR	1						
IS	0.44(**)	1					
PG	0.27(**)	0.53(**)	1				
RL	0.11	0.16	.04	1			
GA	0.37(**)	0.30(**)	0.45(**)	0.01	1		
Extent	0.25(**)	0.35(**)	0.28(**)	0.07	0.24(**)	1	
TE	0.29(**)	0.54(**)	0.41(**)	−0.03	0.25(**)	0.18	1

Notes: SR – supplier R&D resources; IS – information sharing; PG – task programmability; RL – relationship length; GA – goal congruence; Extent – Extent of involvement; TE – product development team effectiveness.

**Correlation is significant at $p < 0.01$.

In order to distinguish the resource effect from the agency effect, we used Hierarchical Regression Analysis (HRA) that allows combining variables into blocks and adding one block each time to the regression model. HRA better serves the purpose of this study than other complicated techniques like SEM as it could easily handle the sequential introduction of variables. The feature of this model is the combination of two theories. Since the resource side has been well understood, the resource factor was first introduced as independent variable in the model. Then agency factors were added into the model as a separate independent variable block. The dominant effect of the two factors became evident when tested in a sequential manner. In addition, HRA was chosen over stepwise regression because HRA is more a theory-driven testing technique where researchers must tell the statistical software what order to put the predictor variables into the regression equation. The order in which the predictor variables are entered is not a statistical decision, but a theory on which the hypothesis is based.

We present results of three models to facilitate the assessment of the relative explanatory power of each or each set of factors. In the first regression model, we regressed project team effectiveness only on the extent of involvement, the control variable. In the second regression model, we regressed project team effectiveness on both supplier's R&D resources and the control variable. The result indicates how much effectiveness the product development team could benefit from the resource perspective. In the third regression model, we regressed project team effectiveness on supplier's R&D resources, the control variable as well as the five agency factors. Since the three models were nested, we compared the change in R^2 and the associated change of F statistics to assess whether the relatively complex model can significantly improve the goodness fit of the baseline model.

Table 3 reports the regression results. The impact from the control variable is marginally significant in the first model ($p = 0.051$). The resource factor (i.e. supplier's R&D resources) was found to produce significant impact on project team effectiveness in the second model, improving the explanatory power to about 10% of the total variance. However, when the agency factors were added into the third model, the resource factor turned out to be insignificant whereas the impact of two agency factors was quite substantial. In addition, the change of F statistics was significant within the third model explaining about 43% of the variance of project team effectiveness. Looking at the results together, we concluded that H_{2a} and H_{2c} are supported whereas H_1 , H_{2b} , H_{2d} are not. Since the impact of the extent of involvement was never statistically significant, the potential interaction effect will be marginal, if any. Therefore, it was not explored further.

Table 3 Regression results

<i>Independent variables</i>	<i>Model 1</i>	<i>Model 2</i>	<i>Model 3</i>
Extent of involvement	0.21 [*]	0.16	-0.03
Supplier's R&D resources		0.22 [*]	0.01
Information sharing			0.33 ^{**}
Goal congruence			0.01
Task programmability			0.44 ^{**}
Relationship length			0.08
R^2	0.04	0.09	0.43
ΔR^2	0.04	0.09	0.34
ΔF	3.92	4.21	12.63
<i>p-value</i>	0.051	0.018	<0.001

Notes: ^{*} $p = 0.051$; ^{*} $p < 0.05$; ^{**} $p < 0.01$.

6 Discussions and implications

Overall, the results support the dominant effect of the agency factors over the resource factor on project team effectiveness. The results indicate that finding the right partner with complementary resources alone is insufficient to explain the success or failure of ESI. Rather, the interaction of the manufacturer and supplier and the management of

the on-going relationship play more critical roles. The empirical results highlight the importance of framing the dynamic relationship of the manufacturer and supplier in the collaborative design process using the agency theory. Yet, it should be noted that not all the agency factors are equally important. In particular, this study identifies two key agency factors (i.e. information sharing and task programmability) to explain the success of ESI implementation beyond the selection of the appropriate supplier.

Many scholars have already emphasised the importance of information exchange between the two parties in ESI process (Brown and Eisenhardt, 1995; Dowlatshahi, 1997; Humphreys, 2004; Lyu and Chang, 2007). The underlying rationale is that in a collaborative relationship, communication among project team and with outsiders increases the amount and variety of information that stimulates the performance of development teams.

This study provides additional support for information sharing from a different perspective. Information sharing between the manufacturer and supplier not only ensures that NPD information is transparent and secure but also serves as a safeguarding mechanism. If the manufacturer shares its knowledge of customers' requirements and expectations as well as its target cost with the supplier, the supplier can suggest feasible designs and accelerate the decision-making process. In addition, once the manufacturer is open to discuss these issues, the supplier feels trusted and engaged. The supplier then becomes more motivated and committed to the project and has little incentive to reduce its effort. In reverse, if the supplier communicates with the manufacturer about its cost structure and technology capability, asymmetric information is reduced as well. Consequently, it is relatively easier for the manufacturer to monitor and evaluate the supplier's behaviour and performance. This also prevents the supplier from 'shrinking'. Through this governance structure, the manufacturer can effectively monitor the supplier's behaviour and align the interests of both parties.

Task programmability offers another way to reduce asymmetric information. If the responsibilities and activities for each party can be clearly specified, the supplier's effort or behaviour can be easily and readily disclosed. Apparently, new product development is an innovation process where the design may be changed and adjusted many times before it is finalised. It is extremely difficult, if not impossible, to outline the detailed activities in the whole process *ex ante*. However, if the development process can be broken down into many stages and in each stage the objective is specific and responsibilities of each party are clearly divided, the manufacturer knows what to expect from the supplier and how to evaluate the supplier's performance for each stage. Under such a situation, it is hard for the supplier to behave opportunistically.

The insignificant factors are goal congruence and relationship length. At first glance this appears surprising; however, on closer examination it becomes understandable when one examines the relationships among the agency factors. Conceptually, it can be argued that information sharing serves as a means to achieve goal congruence between the two parties and better alignment of their interests as previously discussed. Empirically, this argument can be verified from the significant correlation between information sharing and goal congruence (0.30, $p < 0.001$). Therefore, goal congruence can be viewed as a result of this fundamental governance structure; and its impact on team effectiveness could be absorbed already into information sharing.

The counter-intuitive results of the relationship length can be further explained from two aspects. First, ESI is a relatively new practice compared with quality management practice and most of the firms are at the early stage of supplier involvement. Put alternatively, fewer firms have repetitive relationships with a supplier or have collaborated with that

supplier for more than one project. This can be confirmed by the mean values of the measurement for relationship length (Table 1), which are higher than the measurement for other constructs, indicating that respondents tend to disagree with the statement. Even though the theory predicts that the positive impact of longer relationship on project team effectiveness, its impact may not yet be realised at such an early stage. Second, the uncertain and risky nature of the new product development process does not necessarily support the longer repetitive relationships between manufacturers and suppliers. The influential work by Granovetter (1973) ('The strength of weak tie') indicates that weak ties in the supply chain network design promote innovation. Though agency theory suggests longer contractual relationship (i.e. strong tie) could serve as a governance structure to facilitate the governance and reduce goal misalignment, it may not be a 'realistic' means to achieve the objective of supplier involvement. To take maximum advantage of the supplier's knowledge and capability, the manufacturer needs to keep updating its supply chain network. Often times this means reaching beyond the existing network to use some unfamiliar suppliers but with a resource advantage. We suspect that these two effects cancel out each other and leave a marginal impact on product team effectiveness.

7 Conclusions, limitations and future research

The major findings of this study enrich the understanding of ESI by offering a possible explanation of why some firms succeed in implementing ESI while others fail. We provide this understanding by using a combination of the resource-based view and the agency theory. Though prior studies of early supplier involvement have evolved under different theoretical premises independently, this study is the first one to integrate them and empirically test the overall effect of using suppliers on product team effectiveness in the product design process. In addition, past research has paid much attention to selecting the right suppliers and emphasised the collaboration effect, but neglected the dynamic relationship between the manufacturer and supplier during the collaboration process. The results of this study demonstrate that the supplier's R&D resources are necessary but insufficient for a successful collaboration. A successful collaboration requires the manufacturer's constant attention and effort to manage the on-going relationship. The study also extends the agency theory to the collaborative relationship when engaging the supplier in NPD. To improve product team effectiveness and eventually increase the likelihood of innovation, the study validates the importance for the manufacturer of developing governance structures to curb the supplier's opportunistic behaviour.

In particular, this study empirically tested a set of agency factors and identified the key governance structures. The key governance structures are quite robust regardless of the extent of involvement. The results of this study provide critical managerial implications for firms considering ESI practice in innovation processes. Involving suppliers in product development brings the manufacturer opportunities to tap into the supplier's capability, but also exposes it to various kinds of risks and associated costs. To deal with the risks and maximise the benefit of ESI, this study suggests investing in information systems and strengthening communication as well as detailing activities and responsibilities of each party in every stage of the development process. The goal of these efforts is to limit the supplier's potential self-interest behaviour and consequently improve the quality of collaboration.

The article acknowledges a number of limitations of this study. First, we used a secondary survey data; and the size and the content of the dataset do not offer us the ability to test the full model with all the agency variables. Still, the results of this study provide strong empirical evidence that managing the dynamic relationships and preventing the opportunistic behaviour are critical for high-quality collaboration. The results, undoubtedly, will encourage more future research to focus on the governance structure of collaborative relationship.

Second, the measurement instrument is not optimal even though the reliability and validity of all constructs are satisfactory. In terms of 'supplier's R&D resources', we did not have direct measurement in the survey instrument and had to use proxies. The respondents were asked to rate the importance of several factors (Table 1) when selecting a specific supplier. If the manufacturer could find the best supplier that meets these selection criteria, the importance of the rating also defines the traits of selected suppliers. Fortunately, 72% of the respondents have more than one supplier available who met minimum requirements on all relevant resource dimensions and a competitive analysis of different suppliers was carried out to select the 'best' suppliers. This affirmed our confidence in use of the proxy measures to indirectly assess whether the supplier possesses the appropriate resources or knowledge that the manufacturer requires.

In spite of the limitations, this study provides many opportunities for future research. First, only internal mechanisms were considered to counteract the supplier's potential opportunistic behaviour. In fact, the manufacturer could also take advantage of external forces. For example, the manufacturer could involve more than one supplier for a specific part in the product development process. The competition for future business with the manufacturer could potentially drive away the suppliers' opportunistic behaviour. This practice has been adopted in the automobile industry. An interesting question is to assess the effectiveness of the external vis-à-vis internal governance mechanism on product team effectiveness.

Second, to motivate the supplier's effort, it may be valuable to break the whole development project into multiple stages. At the end of each stage the manufacturer has the opportunity to evaluate the supplier's behaviour and decide whether to hand over the development completely to the supplier or take it over and develop in-house. Theoretically, the more stages the manufacturer creates, the higher the benefits obtained. On the other hand, it also threatens the basic trust between the two parties and reduces the morale of the supplier in the contribution to the project. As such, how to manage the stages to maximise the value and minimise the risk warrants further investigation.

Third, given the potential conflict between the manufacturer and supplier, it may be beneficial for the manufacturer to keep certain R&D capabilities for the major parts/sub-assembly systems. The value can be viewed from the real options theory. A real option is the investment in physical and human assets (rather than financial instrument) that provides the opportunity to respond to future uncertain events (Kogut and Kulatilaka, 2001). The manufacturer's investment in R&D capabilities for major parts can be viewed as a real option. By holding this option, the manufacturer minimises its downside risk (i.e. it always has the option of handling the innovation by itself if the agency cost is too substantial) while its potential benefit could be unlimited (i.e. the collaboration works well and generates great success for the new product). The key question is under what situation the manufacturer should consider the real options and under what conditions it should exercise the options.

In spite of reducing asymmetric information, Eisenhardt (1989) argued that the other option is to contract on the outcomes of the agent's behaviour. Though writing a complete contract is impossible in ESI due to the high uncertainty associated with the development process, certain outcome-based rewards and incentives may still work well in aligning the interests. The goal is again to motivate the agent by co-aligning the agent's preferences with those of the principal. If the supplier has a firm commitment from the manufacturer to supply this item for the life of the product, it has no pressure to work out the best design and may not place it on the top of its priority list in resource allocation. In contrast, if the supplier's future business with the manufacturer is based on the success of this co-development effort, the supplier has to do its best to generate a quality product and get it to the market as soon as possible in order to capture the future benefit. Therefore, we also suggest future studies focusing on developing contracts or incentive schemes to overcome the agency problem.

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